NLP1

Introduction

Wilker Aziz



https://nlp-illc.github.io/nlp1/2026/

Fall 2026

Outline

1. Course Organisation

2. What is NLP?

Course Organisation

Team

Coordinators: Wilker Aziz and Martha Lewis

Senior staff: Fivos Tzavellos

Lab coordinators: Vera, Evgenia and Milan

TAs: Ivo, John , Luan, Alexandru, Ippokratis, Lisanne, Lukasz, Mattia, Naval, Oscar and Xanthos.

see Canvas for contact information

Platforms

- Class and lab materials: `https://nlp-illc.github.io/nlp1/`
- Schedule: datanose
- Submitting a deliverable: Canvas/Assignments
- Forming groups: Canvas/People/Groups
- Async Q&A: `https://edstem.org/eu/courses/3201/discussion`
- ANS: exam grade, feedback and discussion
- You will also find in Canvas
  - class recordings
  - complete info on assessment and policies
  - announcements

Text book. We often refer to chapters of Jurafsky: `https://web.stanford.edu/~jurafsky/slp3/`

Learning activities

- Lectures (~ 4h/week): core material (or ‘theory’)
- Werkcolleges (~ 4h/week):
  - practical NLP experience
  - bridging ‘theory’ and ‘practice’ (can be very useful for better understanding the core material)
- Self-study (~ 12h/week): an indispensable activity that we hope you won’t neglect; examples
  - reading and reflecting upon textbook chapters
  - watching a pre-recorded video and exercising a skill
  - familiarising yourself with tutorials
  - developing projects

Assessment

Components

- **E**xam (week 8): 80%
- 3 mini **P**rojects (assessed in weeks 3, 5 and 7): 15%
- 1 **B**log post (week 7): 5%

Grade: SIS($0.8E + 0.05P_1 + 0.05P_2 + 0.05P_3 + 0.05B$)

- No requirements on the components
- You pass the course if your grade ≥ 6 after SIS-rounding
- The resit grade substitute the entire grade

We have three tutorials:

- T1 (week 1): text segmentation and text classification
- T2 (week 3): compositional text encoders
- T3 (week 5): LLM-based NLP

Each tutorial then feeds into a mini project:

- P1 (week 2; assessment week 3)
- P2 (week 4; assessment week 5)
- P3 (week 6; assessment week 7)

We assess your understanding of each T by interviewing you about the corresponding P:

- Oral assessment: 15 min (strict).
- Deliverables: a handful of slides and URL to proj repo.

Tasks and Datasets

You will choose a task/dataset and work on that task throughout all practicals.

Here's a list with suggestions:
`https://nlp-illc.github.io/nlp1/2026/datasets/`

Datasets outside this list are allowed, but you need to discuss it with your TA and they need to approve it (this is to help avoid choices that turn out frustrating down the road).

Blog Post

You then write a blog post, a short piece of scientific/research dissemination (1500 words), highlighting a remarkable aspect of your projects.

If you consent, the blog post will be published in our course's blog: `https://nlp-illc.github.io/nlp1/blog/`

- you will get support/advice (in week 7);
- and feedback and grade from your TA (later in the block);
- we will keep our blog live for years to come, so you can always include this post amongst your research output;
- publicising it or not does not affect your grade.

Exam Preparation

We will share exam-like questions throughout the course as part of class materials and, in due time, we will share a practice exam.

1. How language works:
   what language is and how it works

1. How language works:
   what language is and how it works
2. The building blocks of NLU and NLG:
   text classification | text generation

1. How language works:
   what language is and how it works
2. The building blocks of NLU and NLG:
   text classification | text generation
3. Meaning:
   distributional semantics | compositionality

1. How language works:
   what language is and how it works
2. The building blocks of NLU and NLG:
   text classification | text generation
3. Meaning:
   distributional semantics | compositionality
4. Discourse:
   document representations | coreference

1. How language works:
   what language is and how it works
2. The building blocks of NLU and NLG:
   text classification | text generation
3. Meaning:
   distributional semantics | compositionality
4. Discourse:
   document representations | coreference
5. The building blocks of modern NLP:
   sequence modelling | transfer learning

1. How language works:
   what language is and how it works
2. The building blocks of NLU and NLG:
   text classification | text generation
3. Meaning:
   distributional semantics | compositionality
4. Discourse:
   document representations | coreference
5. The building blocks of modern NLP:
   sequence modelling | transfer learning
6. Large language models:
   training, adaptation and applications | post-training

1. How language works:
   what language is and how it works
2. The building blocks of NLU and NLG:
   text classification | text generation
3. Meaning:
   distributional semantics | compositionality
4. Discourse:
   document representations | coreference
5. The building blocks of modern NLP:
   sequence modelling | transfer learning
6. Large language models:
   training, adaptation and applications | post-training
7. Looking inside and looking ahead:
   interpretability | broader impact

Action Points

- By Nov 2, fill in the consent form for recordings (see Canvas/Quizzes);
- By Nov 2, form your group in Canvas (see Canvas/People/Groups)
  You can only sign up with people from your own Werkcollege group; if you need to swap groups then, you have to contact Fivos with the person you are swapping with in the cc.

After Nov 2, it will be possible to sign up to oral examination slots (an announcement will follow).

What is NLP?

What is NLP?

Natural language processing concerns the science and engineering behind **natural language understanding** (NLU) and **natural language generation** (NLG) by computers.

What is NLP?

Natural language processing concerns the science and engineering behind **natural language understanding** (NLU) and **natural language generation** (NLG) by computers.

Nowadays I barely need to introduce NLP, you all know large language models (LLMs), chat bots, coding assistants, and all the language understanding and generation that we routinely associate with modern AI.

Historical Progress in NLP

Georgetown–IBM (1954): a carefully prepared demonstration of Russian-English translation using a selected vocabulary and a few grammar rules.

Anyone with access to the internet (today): language models 'understand' flexible instructions, 'take actions' (such as search the web) and 'respond' in ordinary language.

Historical Progress in NLP

Georgetown–IBM (1954): a carefully prepared demonstration of Russian-English translation using a selected vocabulary and a few grammar rules.

Anyone with access to the internet (today): language models 'understand' flexible instructions, 'take actions' (such as search the web) and 'respond' in ordinary language.

- Me: "I need a logo for my research unit and a tagline (1 line pitch), the logo should be simple and geometrical (use SVG); we work on trustworthy NLP."
- UvA AI Chat asks me to fill in a form with some detail needed for the best advice possible, it then searches the web, writes and compiles some code (for the logo) and offers three taglines I can choose from.

Of course you want to see the result :)



Suggested tagline: *Understanding language. Building trust.*

The logo concept uses ***two interlocking geometric forms*** *to suggest dialogue and mutual support—connecting NLP with reliability without relying on a generic shield or checkmark. Navy and teal give it a restrained academic identity.*

Two more explicit tagline options:

- *Advancing reliable and trustworthy language technology.*
- *Understanding language for AI we can trust.*

My recommendation is the first: short, memorable, and broad enough to cover both fundamental research and technology development.

Progress in NLP

It's not always amusing. In 2023,[1]

- Steven Schwartz (Lawyer): "Is Varghese a real case?"
- ChatGPT confirms that it is, then claims it can be found in legal research databases.

mr. Schwartz went on to quote the case, **but it did not exist!**

But it can also be truly inspiring:

- Check the demo of DeepMind's ASL translation tool: `https://youtu.be/c84y9gAY90c?t=3125`
- and the research by my colleagues at Amsterdam's SignLab: `https://www.signlab-amsterdam.nl`

[1] `https://www.bbc.com/news/world-us-canada-65735769`

NLP1 at the MSc AI

This course will trace the technical milestones that took us from early attempts at machine translation in 1950s to today's large language models (LLMs).

‘Why bother with the path, when we are here?’

Technology is neither good nor bad; nor is it neutral.

— Melvin Kranzberg

After all,
radical simply means ‘grasping things at the root’.

— Angela Davis

Understanding how we got here helps us distinguish what we have learned about language and computation from the particular ways that knowledge is owned, deployed, and paid for today.

‘Why bother with the path, when we are here?’

The course has a technical focus, but we cannot shy away from technology’s broader impact.

Whose knowledge? Decades of research and public investment.

Whose work and resources? Training data, annotation labour, infrastructure, energy, water, and the distribution of their costs.

Whose choices? Who controls these systems, sets their objectives, and decides which alternatives receive investment?

The path helps us distinguish what we have learned about language and computation from the ways that knowledge is owned, deployed, and paid for today.

What makes NLP so difficult anyway?

A famous philosophical 'dialogue' between empiricism and rationalism can be revealing:

> *"Nothing is in the intellect that was not first in the senses."*
>
> *— (often attributed to) Thomas Aquinas*

What makes NLP so difficult anyway?

A famous philosophical 'dialogue' between empiricism and rationalism can be revealing:

> *"Nothing is in the intellect that was not first in the senses."*
>
> *— (often attributed to) Thomas Aquinas*

To which Leibniz retorted

> *"...except the intellect itself."*
>
> *— Leibniz*

What makes NLP so difficult anyway?

A famous philosophical 'dialogue' between empiricism and rationalism can be revealing:

> *"Nothing is in the intellect that was not first in the senses."*
>
> *— (often attributed to) Thomas Aquinas*

To which Leibniz retorted

> *"...except the intellect itself."*
>
> *— Leibniz*

You and I have a language faculty, once equipped to acquire language (when we were children), now fully equipped to use it. **Computers do not**.

What's Next?

- Today: what language is;
- Next class: how language works.